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# Interventions to prevent or cease electronic cigarette use in children and adolescents (Review)

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[Intervention Review]

# Interventions to prevent or cease electronic cigarette use in children and adolescents

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## ABSTRACT

### Rationale

The prevalence of e-cigarette use has recently increased globally amongst children and adolescents. In response to this increase and emerging evidence about the potential harms of e-cigarettes in children and adolescents, leading public health organisations have called for approaches to address e-cigarette use. Whilst evaluations of approaches to reduce uptake and use regularly appear in the literature, the collective long-term benefit of these is currently unclear.

### Objectives

The co-primary objectives were to: (1) evaluate the effectiveness of interventions to prevent e-cigarette use in children and adolescents (aged 19 years and younger) with no prior use, relative to no intervention, waiting-list control, usual practice, or an alternative intervention; and (2) evaluate the effectiveness of interventions to cease e-cigarette use in children and adolescents (aged 19 years and younger) reporting current use, relative to no intervention, waiting-list control, usual practice, or an alternative intervention. Secondary objectives were to: (1) examine the effect of such interventions on child and adolescent use of other tobacco products (e.g. cigarettes, cigars, chewing tobacco and pouches); and (2) describe the unintended adverse effects of the intervention on individuals, or on organisations where such interventions were being implemented.

### Search methods

We searched CENTRAL, Ovid MEDLINE, Ovid Embase, Ovid PsycINFO, EBSCO CINAHL and Europe PMC on 1st September 2025. Additionally, we searched two trial registry platforms (WHO International Clinical Trials Registry Platform; ClinicalTrials.gov), and reference lists of relevant systematic reviews. We contacted corresponding authors of articles identified as ongoing studies.


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**Eligibility criteria**

We included randomised controlled trials (RCTs), including cluster-RCTs, factorial RCTs, and stepped-wedge RCTs. To be eligible, the primary targets of the interventions were children and adolescents aged 19 years or younger. Interventions could have been conducted in any setting, including the community, school, health services, or the home, and must have sought to influence children or adolescent (or both) e-cigarette use directly. Studies with a comparator of no intervention (i.e. control), waiting-list control, usual practice, or an alternative intervention not targeting e-cigarette use were eligible. Two review authors independently screened the titles and abstracts of references, with any discrepancies resolved through consensus or a third reviewer.

**Outcomes**

The critical review outcome was e-cigarette use in children and adolescents aged 19 years or younger. We included measures to assess the effectiveness of interventions to: prevent child and adolescent e-cigarette use (including measures of e-cigarette use amongst those who were never-users); and cease e-cigarette use (including measures of e-cigarette use amongst children and adolescents who were current e-cigarette users at baseline). We included data measured at least six months post-baseline. Outcomes were current use (defined as use in the past 30 days), ever use (defined as any lifetime use) and adverse effects.

**Risk of bias**

Risk of bias for all included studies was assessed using the Cochrane RoB 2 tool. We applied this tool to the critical and important review outcomes from each included study to assess and rate each outcome as low, some concerns or high for all relevant domains.

**Synthesis methods**

Pairs of review authors independently extracted information from the included studies, with any discrepancies resolved through consensus or a third reviewer when required. Meta-analyses were conducted using a random-effects model where data were suitable for pooling, with two prevention studies measuring e-cigarette use pooled. Studies and outcomes unsuitable for pooling in meta-analyses were instead summarised narratively.

**Included studies**

We identified three studies with 10,510 participants as eligible for inclusion. A further 30 studies were identified as ongoing and five are awaiting classification and likely to be eligible for a future update. Two studies employed a cluster-RCT design to test the effectiveness of school-based interventions to prevent adolescent e-cigarette use, with one study judged to have ‘some concerns’ for the overall risk of bias for the e-cigarette ever-use outcome and the other study judged as 'high risk'. The remaining study employed an RCT design to test the effectiveness of a community-based intervention (delivered via text-messages) to support adolescents to cease e-cigarette use. We judged the overall risk of bias as low for this study which reported an e-cigarette current-use outcome. Included studies were conducted in the United States, Australia and Sweden.

**Synthesis of results**

For the critical outcome of e-cigarette ever use, a meta-analysis of two studies (one high risk of bias, one some concerns) found that prevention interventions may prevent ever use, relative to usual care, although evidence is very uncertain (RR 0.94, 95% CI 0.89 to 0.99; 2 studies; 5306 participants; random-effects model; very-low certainty evidence). The certainty of evidence in this effect was downgraded due to risk of bias, indirectness and imprecision. One study reporting a prevention intervention reported no adverse effects. For the critical outcome of e-cigarette current use, one study reported that a cessation-focused intervention is likely to reduce adolescent current use of e-cigarettes (RR 0.73, 95% CI 0.65 to 0.82; 1 study, 1064 participants; moderate-certainty evidence). The certainty of evidence in this effect was downgraded due to indirectness. The cessation intervention did not report on adverse effects of the intervention.

**Authors' conclusions**

Given only three randomised studies were included in the review, there is limited evidence, of very low-to-moderate-certainty, that interventions may be effective in preventing or ceasing adolescent e-cigarette use. As findings of the 30 ongoing studies are published, certainty of evidence of effects may improve. Until then, the findings of this review should be considered together with evidence from studies employing other trial designs not eligible for inclusion in this review to guide actions to prevent or cease e-cigarette use.

This is a living systematic review. We search for new evidence every month and update the review when we identify relevant new evidence.

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**Registration**

This review is registered in the Cochrane Database of Systematic Reviews. The protocol (https://doi.org/10.1002/14651858.CD015511) and previous version of the review (https://doi.org/10.1002/14651858.CD015511.pub2) are published in the Cochrane Library.


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## PLAIN LANGUAGE SUMMARY

### What are the most effective ways to prevent children and adolescents from using electronic cigarettes?

### Key messages

- School-based programmes may prevent adolescents from starting to use electronic cigarettes (e-cigarettes) and a community-based programme delivered by text message probably helps adolescents quit using e-cigarettes.
- The evidence is very limited because only three studies contributed to the results.
- Another 30 studies are underway, and we have five more published studies to check. They may provide more answers in the future.

### What are electronic cigarettes?

E-cigarettes, also known as vapes, are battery-operated devices that heat a liquid and produce an aerosol (a mix of small particles released in the air) which is inhaled through the mouth into the lungs. They can include electronic cigars, 'snus' (a Swedish tobacco device) or water pipes, and may or may not contain nicotine.

It was generally thought that these devices might lead to people stopping smoking ('current-users') and it is clear that they help adults to quit. However, young people who are not smoking ('never-users') are more likely to try these devices, and they can cause harm to the lungs, such as asthma and effects on development and learning.

### What type of programmes are used to stop children and adolescents from using e-cigarettes?

The types of programmes included: education, training and communication (either community- or school-based) or public announcements; other programmes could include family strategies, medicines or counselling, all designed to influence children or adolescents to prevent or stop e-cigarette use.

### What did we want to find out?

We wanted to find out:

- whether programmes designed to prevent children and adolescents from starting to use e-cigarettes, or to quit using them, are effective;
- if the programmes had any unwanted effects on children's and adolescents' health, or on the organisations that delivered them.

### What did we do?

We searched for studies that investigated programmes to prevent children and adolescents from starting to use e-cigarettes (never-users) or to help them quit using e-cigarettes (current-users).

- Studies could compare programmes with no programme, being on a waiting-list for a programme or another general health programme. The programmes had to be delivered in the community, school, health services, hospitals, or the home.
- People had to be children or adolescents aged 19 years or younger, but also included their parents or caregivers and health professionals.
- The use of tobacco was measured either in the laboratory or by parent- or self-reports, questionnaires or surveys.

We compared and summarised the results, and rated our confidence in the evidence, based on factors such as study methods and sizes.

### What did we find?

We found three studies with 10,510 adolescents.

- All participants in the studies were adolescents, with an age range from 12 to 17 years.
- Two of the studies (undertaken in Sweden and Australia) were delivered in schools. The Swedish study (1176 adolescents) lasted three years and required an agreement signed between caregivers or parents and the adolescent pledging to not take up smoking; they also received an annual education session. This programme was compared with four annual educational sessions. The Australian study (7654 adolescents) had four weeks of programmes (40-minute lessons) and was compared with the usual health curriculum.
- One US prevention study (with 1681 current-users), delivered in the community, used a text-based message system which lasted four or 14 weeks (depending on whether a quit date was given or no quit date was given, respectively). It was compared with either monthly text messages that reported e-cigarette use or a waiting-list control (that is, people on a waiting list to receive treatment).

### Main results


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- The two combined studies found that school-based programmes may help prevent adolescents from ever trying e-cigarettes, but we are very uncertain about the results: 31% of adolescents on school programmes may be prevented from using e-cigarettes compared to 33% of those on an alternative programme (two studies, 8830 people).
- The remaining study found that a community-based programme (for current-users) probably helps adolescents quit e-cigarette use: 45% of adolescents on the programme reported continuing the use of e-cigarettes compared to 62% with an alternative programme or a waiting-list (1 study, 1681 people).
- No studies provided information on the unwanted effects of the programmes.

**What are the limitations of this evidence?**

Given that the review only included three studies, very limited evidence is available for us to determine what programmes may work to help prevent or stop adolescent e-cigarette use. Our confidence in the evidence was very low-to-moderate because there were very few studies found, the programmes were different, and the studies lasted for different time periods.

**How up-to-date is this evidence?**

This evidence is current to September 1, 2025. This updates our previous review published in 2024.


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## SUMMARY OF FINDINGS

### Summary of findings 1. Summary of findings table - Interventions to prevent electronic cigarette use in children and adolescents

**Interventions to prevent electronic cigarette use in children and adolescents**

**Patient or population:** children and adolescents
**Setting:** school-based
**Intervention:** interventions designed to prevent electronic cigarette use
**Comparison:** minimal intervention or active control

| Outcomes | Anticipated absolute effects* (95% CI) | | Relative effect (95% CI) | № of participants (studies) | Certainty of the evidence (GRADE) | Comments |
|---|---|---|---|---|---|---|
| | Risk with minimal intervention or active control | Risk with interventions designed to prevent electronic cigarette use | | | | |
| E-cigarette ever-use<br>follow-up: range 12 months to 18 months | 326 per 1000 | **307 per 1000**<br>(291 to 323) | **RR 0.94**<br>(0.89 to 0.99) | 5306<br>(2 RCTs) | ⊕⊖⊖⊖<br>Very low[a,b,c] | |
| E-cigarette current-use<br>follow-up: 18 months | 22 per 1000 | **9 per 1000**<br>(3 to 23) | **RR 0.40**<br>(0.15 to 1.07) | 3949<br>(1 RCT) | ⊕⊖⊖⊖<br>Very low[a,b,c] | |
| Unintended adverse consequences of the intervention - not reported | - | - | - | - | - | No studies reported data on unintended adverse consequences. |

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

**CI:** confidence interval; **RR:** risk ratio

**GRADE Working Group grades of evidence**
**High certainty:** we are very confident that the true effect lies close to that of the estimate of the effect.
**Moderate certainty:** we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
**Low certainty:** our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
**Very low certainty:** we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_455907313968586534.


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[a] Downgraded by one level for risk of bias
[b] Downgraded by one level for indirectness
[c] Downgraded by one level for imprecision

**Summary of findings 2. Summary of findings table - Interventions to cease electronic cigarette use in children and adolescents**

**Interventions to cease electronic cigarette use in children and adolescents**

**Patient or population:** children and adolescents
**Setting:** community-based
**Intervention:** interventions designed to cease electronic cigarette use
**Comparison:** assessment-only control

| Outcomes | Anticipated absolute effects* (95% CI) | | Relative effect (95% CI) | № of participants (studies) | Certainty of the evidence (GRADE) | Comments |
|---|---|---|---|---|---|---|
| | Risk with assessment-only control | Risk with interventions designed to cease electronic cigarette use | | | | |
| E-cigarette current use follow-up: 7 months | 617 per 1000 | **450 per 1000** (401 to 506) | **RR 0.73** (0.65 to 0.82) | 1064 (1 RCT) | ⊕⊕⊕⊖ Moderate[a] | Derived from 30-day point prevalence abstinence from e-cigarettes. |
| Unintended adverse consequences of the intervention - not reported | - | - | - | - | - | No studies reported data on unintended adverse consequences. |

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

**CI:** confidence interval; **RR:** risk ratio

**GRADE Working Group grades of evidence**
**High certainty:** we are very confident that the true effect lies close to that of the estimate of the effect.
**Moderate certainty:** we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
**Low certainty:** our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
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[a] Downgraded by one level for indirectness.


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# BACKGROUND

## Description of the condition

Electronic nicotine delivery systems (ENDS) and electronic non-nicotine delivery systems (ENNDS) heat a liquid to produce an aerosol to inhale (often known as 'vaping'). The e-liquids typically contain propylene glycol, vegetable glycerine/glycerol, flavour additives, known toxic chemicals and, commonly, nicotine (i.e. ENDS) [1][2]. ENDS and ENNDS come in various forms, including e-cigarettes, e-hookahs, e-cigars, and e-pipes, with e-cigarettes being the most common [3].

Global surveillance data demonstrating e-cigarette use, including dual use of ENDS and ENNDS amongst adolescents, has increased substantially over time across multiple regions [4, 5]. A systematic review synthesising the prevalence of e-cigarette use in children and adolescents (aged 8 to 20 years) from 69 countries and territories reported a pooled prevalence for ever use (defined as any lifetime use) of 17.2% and current use (defined as use in the past 30 days) of 7.8% [6]. Of growing concern is the high prevalence of e-cigarette use amongst adolescents who are otherwise not current or ever-users of tobacco. For example, in one Australian survey of adolescents aged 14 to 17 years, 29.9% of adolescents reported e-cigarette ever use (15.7% in the past month), more than two-thirds (69%) of whom had never smoked tobacco [7]. The increasing use of e-cigarettes is likely influenced by a number of complex interpersonal (e.g. lower perceived risks and harms regarding e-cigarette use) [8], societal (e.g. peer and sibling influences) [9, 10], and environmental (e.g. accessibility and exposure to e-cigarette-related marketing) factors [11].

There remains debate whether the introduction of e-cigarettes has facilitated a decline in tobacco smoking, and if their benefit as a smoking cessation aid for established adult smokers outweighs their potential risk of increasing e-cigarette and tobacco use in children and adolescents [12, 13]. Findings from one Cochrane living systematic review reported high-certainty evidence that e-cigarettes containing nicotine increase tobacco smoking quit rates at six+ months in adult smokers compared to nicotine replacement therapy, and moderate-certainty evidence that they increase quit rates compared to nicotine-free e-cigarettes [14]. Further, a recent Cochrane review reported very low-certainty evidence suggesting that youth e-cigarette use and availability were inversely associated with youth tobacco smoking (i.e. as e-cigarette availability increased, youth smoking rates decreased or, conversely, as e-cigarettes were restricted, youth smoking rates increased) [15]. However, meta-analyses of prospective cohort studies report that e-cigarette use is independently associated with increased risk of initiating cigarette smoking [16, 17], as well as increased inflammatory responses and harmful effects on respiratory outcomes [18].

Reviews of global evidence have also suggested that the use of nicotine-containing e-cigarettes (i.e. ENDS) as well as e-cigarettes marketed as nicotine-free (i.e. ENNDS) poses serious adverse effects to users [19]. These potential harms include respiratory diseases, poisoning, burns, immediate toxicity through inhalation, and seizures [19]. ENNDS devices can often contain cannabis and its associated derivatives such as tetrahydrocannabinol (THC) and cannabidiol (CBD), which are associated with their own health risks, including anxiety and depression [20]. Less serious adverse effects include nausea and throat irritation [19]. Specifically for children and adolescents, systematic review evidence indicates the use of e-cigarettes is significantly associated with respiratory conditions, including asthma [21]. Further, the potential risk to brain development, attention, mental health and learning in adolescents as a result of consuming nicotine, which is commonly found in e-cigarettes, is of particular concern [22, 23, 24].

## Description of the intervention and how it might work

In response to emerging evidence about the potential harmful health effects of e-cigarettes on children and adolescents and the lack of benefits for non-smoking youths, leading public health organisations including the World Health Organization (WHO) have developed recommendations to address this potential public health issue [2]. For example, the WHO Framework Convention on Tobacco Control (FCTC) developed core demand reduction provisions, including price, tax, and non-price measures to reduce the demand for tobacco [25], which have been applied to e-cigarettes [26]. These provisions are part of the MPOWER measures, a set of six cost-effective and high-impact tobacco and e-cigarette control measures introduced by WHO to assist in the country-level implementation of effective interventions to reduce the demand for tobacco and e-cigarettes [27]. The six measures include: (M) monitoring tobacco use and prevention policies; (P) protecting people from tobacco smoke; (O) offering help to quit smoking; (W) warning about the dangers of tobacco; (E) enforcing bans on tobacco advertising, promotion, and sponsorship; and (R) raising taxes on tobacco.

In response to the MPOWER measures introduced by WHO, several countries have implemented national and state-level policy and legislative changes to restrict the supply of e-cigarettes to children and adolescents [2]. For example, the United States (US) passed federal legislation in 2019 which prohibits the sale of all tobacco products, including e-cigarettes, to individuals under 21 years of age. In Australia, in addition to state-level restrictions on underage sales of e-cigarettes that were introduced over the past decade [28], adults aged 18 years are able to purchase vaping products through pharmacies and retail sales in other outlets are prohibited under national legislation [29]. To address the MPOWER non-price measures, national health agencies have recommended the development of education, communication, training, and health promotion programmes targeting children and adolescents to prevent or cease (or both) e-cigarette use [30, 31]. For example, the US Substance Abuse and Mental Health Services Administration has produced an evidence-based resource guide to translate e-cigarette research into recommendations for practice, and to provide examples of how to implement these recommendations via individual, education, and community-level interventions [30]. In recent years, various health promotion programmes have been developed and delivered in different settings to prevent and/or cease e-cigarette use in children and adolescents [32, 33, 34, 35, 36, 37].

Evaluation of public and community-based interventions, delivered via social media, text messages, and online modalities, suggests that these approaches are potentially effective in reducing adolescent e-cigarette use and increasing abstinence rates [33, 34, 35, 37]. School-based programmes targeting school policy and curriculum changes regarding the use of e-cigarettes in schools, in combination with targeting students via social media and online programmes, have shown promise in increasing adolescent knowledge of the potential harms and reducing adolescent intent


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to try e-cigarettes [32, 36, 38]. Although early non-controlled evidence suggests such programmes may be beneficial, few rigorous evaluations have been conducted.

Several theories have been used to describe how the proposed interventions may work to influence child and adolescent health behaviours [39]. However, given that child and adolescent e-cigarette use is a relatively new public health issue, theories to describe how interventions may work to specifically influence child and adolescent e-cigarette use are scarce. Nonetheless, multiple psychosocial theories, including the theory of planned behaviour [40], socio-ecological model [41], and social cognitive theory [42], have been previously used to describe potential causal pathways to child and adolescent tobacco use, and may also be relevant for behaviour related to the initiation and use of e-cigarettes.

Collectively, these theories suggest that an individual's behaviour is influenced by a combination of personal and environmental factors, and that an individual's behaviour is directly related to the context of social interactions and experiences [40, 41, 42, 43]. As such, strategies effective in influencing child and adolescent tobacco use are likely to target both the individual's personal influences (e.g. knowledge, self-efficacy), their parents and carers (as key social influences), and environmental characteristics (such as friendship groups and perceived social norms) ([40, 41, 42]). Nonetheless, the factors targeted by interventions to modify adolescent behaviour will depend on whether such interventions seek to prevent initiation of e-cigarettes (primary prevention) or support their cessation (secondary prevention), as the motivators for use differ. For example, the motivations for initial experimentation with e-cigarettes in adolescents are likely to differ from those who are seeking to quit [44, 45, 46]. Similarly, behaviours related to the cessation of e-cigarettes may be more challenging amongst regular users of ENDS experiencing nicotine addiction, compared to regular users of ENNDS, where this may not be the case [47]. Whilst we anticipate that the majority of interventions aiming to prevent or cease child and adolescent e-cigarette use will be multicomponent and universal, effective interventions will need to target the specific antecedents of these behaviours.

### Why it is important to do this review

Despite the rapidly increasing prevalence in young people, the strength of evidence supporting programmes to prevent or cease (or both) e-cigarette use in children and adolescents is unknown [48]. Whilst no published studies were eligible for inclusion in the previous version of this review, the considerable number of studies identified as ongoing (n = 22) suggests a rapidly emerging evidence-base. In this context and as a public health priority, we are undertaking a living systematic review of this evidence to inform ongoing decision-making. Living systematic reviews are underpinned by continual, active monitoring of the evidence, and they immediately incorporate any new relevant evidence that is identified [49, 50]. Findings from a living systematic review have the potential to provide policymakers, practitioners, and researchers with timely guidance on the types of programmes that should be funded and implemented in order to address this increasingly important public health issue. As such, our living systematic review employing rigorous methodology will provide insight into the effectiveness of current published interventions aiming to prevent or cease child and adolescent e-cigarette use. This is the first update of a Cochrane review first published in 2024.

## OBJECTIVES

The co-primary objectives of the review are to:

- evaluate the effectiveness of interventions to *prevent* e-cigarette use in children and adolescents (aged 19 years and younger) with no prior use, relative to no intervention, waiting-list control, usual practice, or an alternative intervention;
- evaluate the effectiveness of interventions to *cease* e-cigarette use in children and adolescents (aged 19 years and younger) reporting current use, relative to no intervention, waiting-list control, usual practice, or an alternative intervention.

Secondary objectives are to:

- examine the effect of such interventions on child and adolescent use of other tobacco products (e.g. cigarettes, cigars, chewing tobacco and pouches);
- describe the unintended adverse effects of the intervention on individuals (e.g. physical or mental health of individuals), or on organisations (e.g. intervention displacement of key curricula or learning opportunities for school students) where such interventions are being implemented.

## METHODS

The conduct of this review aligns with the Methodological Expectations of Cochrane Intervention Reviews (MECIR) [51] and the reporting is consistent with the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) 2020 [52].

### Differences between protocol and review

This review represents the first update of the living systematic review and was conducted in accordance with the published protocol [53], with the following deviations:

- For the critical and important review outcomes, we amended the hierarchy for outcome selection where multiple validated self-reported measures of the same outcome (i.e. current use) were reported (e.g. repeated point prevalence abstinence (PPA), 30-day PPA, 7-day PPA). We prioritised the 30-day PPA as the primary outcome measure of e-cigarette and current tobacco use. Additional information regarding the hierarchy is provided in the Outcome measures section.
- Minor amendments were made to the Electronic searches due to a lack of access to one database (Clarivate Web of Science Core Collection), which was not searched from January 2025 onwards. From July 2023, we additionally searched EuropePMC to identify preprints.
- The planned synthesis methods included conducting separate meta-analyses for prevention and cessation interventions across multiple outcome measures (e.g. ever use and current use). Due to limited data, only two prevention studies reporting ever use were pooled in a meta-analysis, while the remaining outcomes were summarised narratively. Vote counting approaches were not undertaken due to the limited number of included studies.
- We deviated from the protocol, which specified odds ratios (ORs) as the measure of effect, and instead calculated and reported risk ratios (RRs). This change was made to improve interpretability, as RRs provide a more direct and intuitive estimate of intervention effects and are less likely than ORs to


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overstate associations when outcomes are common, consistent with current Cochrane guidance.

- We planned to calculate missing variance estimates using recommended formulas; however, this was not undertaken. All analyses were based on data as reported in the included studies or provided by study authors upon request.
- Minor wording changes were made to clarify eligibility criteria, update terminology, and align methods with current *Cochrane Handbook* guidance.

## Criteria for considering studies for this review

### Types of studies

We included randomised controlled trials (RCTs), including cluster-RCTs, factorial RCTs, and stepped-wedge RCTs, as these are considered to be the highest-quality study designs to establish intervention effectiveness [54]. Cluster-RCTs were only eligible for inclusion if the trial consisted of a minimum of two intervention sites and two control sites, as per Cochrane Effective Practice of Organisation of Care (EPOC) recommendations [55].

### Types of participants

The primary targets of the interventions were children and adolescents aged 19 years or younger. Participants may have included:

- children and adolescents aged 19 years or younger. Studies including participants older than 19 years were only included if the mean age of the study sample at baseline was 19 years or less; or data for children and adolescents aged 19 years or younger could be extracted separately;
- children and adolescents who were current-users (defined as those that had used e-cigarettes in the past 30 days), ever-users (defined as those that had used e-cigarettes in their lifetime), or never-users (defined as those who had never tried an e-cigarette);
- parents, guardians, and families responsible for the care of children and adolescents aged 19 years or younger;
- professionals within intervention settings (e.g. schools, community, social and welfare settings), including health professionals and teachers, or those in other agencies who may influence e-cigarette use.

### Types of interventions

#### *Intervention*

We anticipate that most e-cigarette interventions evaluated in this review will align with the non-price MPOWER measures introduced by WHO to reduce the demand for tobacco and recently applied to e-cigarettes [27]. These non-price measures include education, communication, training, and public awareness provision [25], including public media campaigns, and community- and school-based education interventions. Nonetheless, we considered any educational, experiential, health promotion, and/or psychological, family, behavioural therapy, pharmacological therapy, counselling, management, structural, policy regulation, or legislative reform interventions amongst current, ever-users, and never-users of e-cigarettes, designed to influence the e-cigarette use of children and adolescents aged 19 years or younger.

Interventions could have been conducted in any setting, including community, school, health services, hospitals, or the home. Included interventions must have sought to influence children and/or adolescents (i.e. individuals aged 19 years or younger) e-cigarette use directly, but could have also sought to influence other risky behaviours such as tobacco use or alcohol consumption. Interventions that target other risk factors were eligible as long as there was a clear component that targeted e-cigarettes. Included interventions could have been single-component or multicomponent (i.e. interventions that included more than one strategy to influence e-cigarette use), and there was no restriction on intervention duration, facilitator, or modality (i.e. the intervention could be delivered by teachers, self-led and delivered face-to-face, online, or by another modality). We excluded interventions undertaken in laboratories or other simulated contexts.

#### *Comparator*

No intervention (i.e. control), waiting-list control, usual practice, or an alternative intervention not targeting e-cigarette use.

## Outcome measures

### Critical outcomes

- **E-cigarette (ENDS or ENNDS) use in children and adolescents aged 19 years or younger**. To match the duality of the co-primary review objectives, we included measures to assess the effectiveness of interventions to:
  - *prevent* child and adolescent e-cigarette use (including measures of e-cigarette use amongst those who were *never-users*) (i.e. those who had never tried an e-cigarette at baseline);
  - *cease* e-cigarette use (including measures of e-cigarette use amongst children and adolescents who were e-cigarette *current-users* at baseline).

Measures of e-cigarette use included:

- current use (most frequently defined as use in the past 30 days);
- ever use (most frequently defined as any lifetime use).

We only included studies with a follow-up assessment of at least three months to account for the potential latency of the behaviour change (i.e. e-cigarette use). E-cigarette use is mostly evaluated using subjective methods (e.g. parent or child/adolescent report), including surveys and questionnaires. To date, there are no biomarkers specific to e-cigarette use given that nicotine may or may not be present or may occur in varying concentrations [56]. If multiple measures of the same e-cigarette outcome were used, we utilised the measure with the greatest evidence of reliability and validity; for example, we included validated measures over those that have not been validated. Where multiple validated self-reported measures of the same e-cigarette outcome were available (e.g. repeated PPA, 30-day PPA, 7-day PPA), we prioritised the 30-day PPA as the primary outcome measure of e-cigarette current use. While repeated PPA can provide a more robust assessment of sustained abstinence, 30-day PPA is more commonly reported, as evidenced by the outcomes specified in the included and ongoing studies in this review and a related prospective meta-analysis [57, 58]. Using this outcome maximises comparability across studies and facilitates potential pooling of data in future meta-analyses.


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**Important outcomes**

- **Child and adolescent use of tobacco**: we extracted data on the impact of the intervention on child and adolescent current or ever use of tobacco. Tobacco use can be biochemically validated using carbon monoxide or cotinine, which we planned to use where available over self-reported tobacco measures. However, since biochemical verification is often considered unnecessary or unfeasible for prevention trials, and as youth tobacco use is often intermittent and concordance has been demonstrated between verified and self-reported measures, we included outcomes assessed using self-report questionnaires or surveys [59], if biochemically validated measures were not available. If multiple measures of the same tobacco outcome were used, we utilised the measure with the greatest evidence of reliability and validity; for example, we included validated measures over those that have not been validated. Where multiple validated self-reported measures of the same tobacco outcome were available (e.g. repeated PPA, 30-day PPA, 7-day PPA), we prioritised 30-day PPA as the primary outcome measure of current tobacco use. While repeated PPA across multiple time points can provide a more robust assessment of sustained tobacco abstinence, 30-day PPA is more commonly reported, as evidenced by the outcomes specified in the included and ongoing studies in this review and a related prospective meta-analysis [57].
- **Unintended adverse consequences of the intervention**: we planned to extract data on any adverse outcomes where these were specified by the included studies. These could have been adverse physical or mental health effects on individuals (e.g. nicotine withdrawal symptoms such as nausea; social connectedness), or the organisations where such interventions are being implemented (e.g. intervention displacement of key curricula or learning opportunities for school students). Such outcomes could be assessed using questionnaires, surveys, or direct observations.

## Search methods for identification of studies

This review represents the first update of the review first published in November 2023 [60].

### Electronic searches

We searched the following electronic databases from 1st May 2023 to 1st September 2025 to identify any eligible trials published since the last published review [60]: Cochrane Central Register of Controlled Trials (CENTRAL) in the Cochrane Library (from inception), Ovid MEDLINE (from 1946), Ovid Embase (from 1947), Ovid PsycINFO (from 1806), and EBSCO Cumulative Index to Nursing and Allied Health Literature (CINAHL) (from 1947) (Supplementary material 1). We adapted elements of the search strategy from a previous review [61], in combination with the sensitivity- and precision-maximising version of the Cochrane RCT filter [62]. This filter was applied to MEDLINE, with the other databases (apart from CENTRAL) using an adapted version. Our search terms included published search filters for "electronic nicotine delivery systems (ENDS) or electronic non-nicotine delivery systems (ENNDS)" AND "study design" AND "children OR adolescents." From August 2023, MESH terms for “child” and “adolescent” were added to the CENTRAL search strategy, and subject headings for “smoking cessation” and “smoking prevention” were incorporated into all database searches.

For the previous version of the review, we also searched Clarivate Web of Science Core Collection (Science Citation Index Expanded and Social Sciences Citation Index) (1947 to December 2024). However, we did not search this database from January 2025 onwards as we no longer had access to the database and determined it did not provide additional unique records in the searches. From July 2023, we additionally searched EuropePMC to identify preprints. We did not impose any restrictions on the language of publication.

#### *Living systematic review considerations*

We will continue to conduct monthly searches of these electronic databases. This review is based on the findings of searches conducted up to September 2025. From April 2026, PubMed was searched monthly for notices of retractions, expressions of concern, or errata in relation to the included studies.

### Searching other resources

We searched for relevant unpublished or grey literature publications in the WHO International Clinical Trials Registry Platform (apps.who.int/trialsearch) and ClinicalTrials.gov (www.clinicaltrials.gov), which collate trial records from multiple national and regional registries worldwide. From June 2024 onwards, ClinicalTrials.gov (www.clinicaltrials.gov) changed their search interface, which led us to substantially change our search strategy (Supplementary material 1). In addition, we handsearched the reference lists of relevant systematic reviews, and contacted authors of relevant protocol papers identified by the electronic database searches.

#### *Living systematic review considerations*

We will continue to conduct monthly searches of trial registries (WHO International Clinical Trials Registry Platform; ClinicalTrials.gov) and search other resources (reference lists of included studies and relevant systematic reviews) every six months. We will continue to contact corresponding authors of ongoing studies as they are identified and will ask them to advise when results are available, or to share early or unpublished data.

The search strategies, including the electronic database and other resources, will be reviewed on a yearly basis to ensure changes to terminology, databases and topic area have been addressed.

## Data collection and analysis

### Selection of studies

Pairs of review authors (for this update: CBa, CBi, SMa, HT, SLY, DDA) independently screened the titles and abstracts of all studies using Covidence software [63]. Prior to commencing screening, review authors independently piloted the abstract and full-text screening strategy with 100 citations and 20 full-text articles to ensure consistency between review authors. Any discrepancies between review authors regarding title and abstract screening were resolved via consensus, or by consulting a third review author (for this update: SMc) when required. No review author was involved in screening their own studies for eligibility.

We obtained the full-text articles of studies that could not clearly be excluded on the basis of title and abstract. Pairs of review authors (for this update: CBa, CBi, SMa, HT, SLY, DDA) independently assessed the full-text articles in Covidence for inclusion in the


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review [63]. Any discrepancies between review authors regarding full-text screening were resolved via consensus, or by consulting a third review author when required (for this update: SMc). We documented the reasons for excluding any full-text manuscripts, with a subset of excluded studies listed in the 'Characteristics of excluded studies' table (Supplementary material 3).

### *Living systematic review considerations*

We will continue to immediately screen any new citations retrieved by the monthly searches of the electronic databases and other resources.

### Data extraction and management

For included studies, pairs of review authors (for this update: CBa, CBi, HT) independently extracted information using a pre-piloted data extraction form developed by the review team. No review author was involved in the extraction of data from their own studies. Any discrepancies regarding data extraction between review authors were resolved via consensus, or by consulting a third review author (for this update: CBa, CBi, HT) when required. We extracted the following data from each included study in sufficient detail to complete a 'Characteristics of included studies' table (Supplementary material 2).

- General information: author name, title, publication date, country, funding source and potential conflicts of interest.
- Study methods: study design, setting, duration, sample size, number of experimental conditions, and number of participants.
- Participant and PROGRESS-Plus characteristics [64]: age, gender, religion, education, race and ethnicity, place of residence, occupation, social capital (e.g. neighbourhood, community, or family support), socioeconomic status, baseline smoking and e-cigarette status (e.g. current, ever).
- Intervention characteristics: intervention description, intensity and length of the intervention, theoretical underpinning, intervention components, and modality (e.g. online, face-to-face).
- Comparator characteristics: comparator description, intensity and length of the comparator, theoretical underpinning, comparator components, and modality (e.g. online, face-to-face).
- Study primary and secondary outcomes (aligned to the review outcomes): outcome definition, data collection method, validity of measures used, effect size and measures of outcome variability.
- Information regarding frequency of e-cigarette use, device type, primary flavour preference, and presence of nicotine within the e-cigarettes.
- Information to enable risk of bias assessment.

We planned to list all treatment arms in the 'Characteristics of included studies' table (Supplementary material 2), even if they are not used in the review.

For ongoing studies and studies awaiting classification, two review authors (for this update: CBi, SMa, HT, DDA) independently extracted information using a pre-piloted data extraction form developed by the review team. Any discrepancies regarding data extraction between the review authors were resolved via consensus, or by consulting a third review author (for this update: CBi, SMa, HT) when required. The following data were extracted and detailed in the 'Characteristics of ongoing studies' tables (Supplementary material 5).

- General information: author name, title, publication date, country.
- Study methods: study design, setting, duration, sample size, number of experimental conditions, and number of participants.
- Participant: age and place of residence.
- Intervention and comparator characteristics: intervention description, components, and modality.
- Study primary and secondary outcomes (aligned to the review outcomes): outcome definition and data collection method.

### Risk of bias assessment in included studies

For all included studies, we assessed the risk of bias using the Cochrane RoB 2 tool [65]. We applied the RoB 2 tool to both critical and important review outcomes from each included study to assess and rate each outcome as low, high, or some concerns for the following risk of bias domains: randomisation process, deviations from the intended intervention, missing outcome data, measurement of the outcome, and selection of the reported result. We assessed cluster-RCTs for an additional risk of bias domain: timing of identification and recruitment of participants. The effect of interest was the effect of assignment to the intervention at baseline (i.e. intention to treat). If included studies reported results for multiple follow-up periods, we assessed risk of bias in the most recent (i.e. longest) study follow-up.

We used the Cochrane RoB 2 guidance document and Excel spreadsheet to implement the tool, in addition to the supporting materials available on the risk of bias website [66]. We determined risk of bias assessments using algorithms in the RoB 2 tool, based on the answers to the signalling questions. Where deemed necessary, we chose to override the algorithm. When deciding the overall risk of bias for a study, we used the recommended criteria below. However, if we considered there was sufficient reason to override the algorithm, we did so and provided a justification.

- Low risk of bias: study is judged at low risk of bias if all domains are at low risk of bias for a given result;
- Some concerns: study is judged as having some concerns if at least one domain for that result is judged as having some concerns, but no domains are classified as high risk of bias for that result;
- High risk of bias: study is judged at high risk of bias for a given result if at least one domain is judged at high risk of bias for that result, or the study is judged as having some concerns for multiple domains for that result.

Two review authors (for this update: SMc, SLY) independently assessed the risk of bias in all studies. No review author was involved in the risk of bias assessments for their own studies included in the review. Any discrepancies between review authors regarding risk of bias assessments were resolved via consensus, or by consulting a third review author when required (for this update: HT, CBa). Risk of bias tables for the co-critical outcomes from individual studies and the risk of bias for the important outcomes are presented in the 'Characteristics of included studies' tables (Supplementary material 2).


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### *Assessment of bias in conducting the systematic review*

We planned to conduct the review according to the published protocol [53], and report any deviations from it in the 'Differences between protocol and review' section of the systematic review.

### Measures of treatment effect

For the co-critical outcomes, we conducted meta-analyses using a random-effects model where appropriate. For prevention interventions, two studies were eligible and were pooled for the outcome of e-cigarette ever use. Planned analyses for other prevention outcomes (e.g. current use) and for cessation interventions could not be undertaken due to insufficient data. Where studies provided unadjusted and adjusted modelled estimates of effects for study outcomes, a hierarchy was employed to guide selection. Specifically, we prioritised unadjusted modelled estimates, due to only RCTs being included and to reduce potential heterogeneity from different adjustments. If unadjusted modelled estimates were not provided, we used adjusted modelled estimates with the least number of adjustments where studies reported multiple analyses with different levels of adjustment. In instances where modelled estimates of the outcome were not provided, the raw data for events and non-events were used to calculate an RR and corresponding 95% CI.

As only dichotomous outcomes were reported by the included studies (e.g. ever and current use), we calculated risk ratios (RRs) with 95% confidence intervals (CIs). Where studies provided odds ratios, these were converted to RRs, using the event rate in the control group as the baseline risk.

### Unit of analysis issues

We examined cluster-RCTs for unit of analysis errors and recorded these if they occurred in the risk of bias tables. As adjusted, modelled effect estimates were reported for the cluster-RCTs, clustering had been appropriately accounted for within the original analyses, and therefore no further adjustments were required.

### Dealing with missing data

We assessed and reported missing data and dropouts in the included studies. All analyses were based on the data as reported in the included studies or additional data provided by study authors (Gardner 2025 [67, 68, 69, 70, 71, 72]) upon request. We did not calculate missing variance estimates or impute missing values. Evidence of potential reporting bias was documented in the risk of bias tables.

### Reporting bias assessment

We were not able to assess possible reporting bias as planned in the review protocol [53] due to the limited number of included studies. In future updates, we will explore reporting bias in any meta-analyses conducted by plotting contour-enhanced funnel plots and visually assessing them for asymmetry and outliers.

### Synthesis methods

One review author (for this update: CBi) entered data into Review Manager Web for analysis [73]. Data entry was checked by a second review author (for this update: HT, CBa). For each critical and important outcome, we conducted meta-analyses using a random-effects model where data were suitable for pooling. We used a random-effects model given the clinical diversity. The Restricted Maximum Likelihood (REML) estimator was used to estimate between-trial variance. The Wald-type method was used to calculate a confidence interval for the meta-analysis effect estimate as there were only a maximum of two studies that contributed to each meta-analysis. Consistent with the co-critical outcomes of the review, we pooled prevention interventions (two studies) and conducted a meta-analysis for the ever-use outcome. For one of these studies, the reported outcome of never use was inverted to calculate ever use. Planned meta-analyses for other prevention outcomes (e.g. current use) and for cessation interventions (one study) could not be undertaken due to insufficient data.

As anticipated, some prevention studies included samples with mixed baseline e-cigarette use status (e.g. current-, ever-, and never-users). However, for both prevention studies, outcome data specific to never-users at baseline could be obtained and thus were included in the review. For cessation studies that included a mixed baseline group (e.g. current- and ever-users), we used effect estimates reported for current-users at baseline.

For the studies and outcomes that were unsuitable for pooling in meta-analyses due to heterogeneity and insufficient data, we summarised findings using narrative synthesis, structured by outcome and intervention type (i.e. prevention or cessation), following guidance in the *Cochrane Handbook for Systematic Reviews of Interventions* [65] and the Synthesis Without Meta-analysis (SWiM) guidelines [74].

### *Living systematic review considerations*

When new evidence (i.e. studies, data or information) meeting the review inclusion criteria is identified, we will continue to immediately extract the data, assess risk of bias, and incorporate new evidence in the synthesis.

### Investigation of heterogeneity and subgroup analysis

Statistical heterogeneity was assessed by visual inspection of forest plots and calculation of the $I^2$ statistic, where we were able to pool data from included studies [65]. We did not perform the planned subgroup analyses previously detailed in the review protocol [53], as there were insufficient studies to estimate effects in various subgroups.

### *Equity-related assessment*

We extracted data on PROGRESS-Plus characteristics where reported in included studies, to explore potential health equity effects [64]. These included age, gender, religion, education, race and ethnicity, place of residence, occupation, social capital and socioeconomic status.

### Sensitivity analysis

We did not perform the sensitivity analyses previously outlined in the review protocol [53], as there were insufficient studies to test the robustness of the effect measures.

### Certainty of the evidence assessment

We used GRADE to assess the overall certainty of the evidence for the critical review outcome of e-cigarette use. We intended to use GRADE to assess unintended adverse consequences of the intervention as per the *Cochrane Handbook* [75], however,


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data were not available for this update. The most recent (i.e. longest) follow-up time point was the time point of interest. The findings of GRADE are presented within the summary of findings tables developed using GRADEpro GDT software [76], which also report for each outcome the treatment effect estimate, number of included studies and participants, and the assessment of the overall certainty of the evidence. Two review authors (for this update: RH, ES, CBa) independently assessed the certainty of the evidence, with any disagreements resolved by consensus, or by consulting a third review author (for this update: HT, CBa), if required.

As per GRADE recommendations, we assessed critical outcome measures against the relevant GRADE criteria to obtain an overall rating and overall level of certainty of the evidence. We considered the following GRADE criteria for lowering the certainty of the evidence.

- Risk of bias (i.e. an outcome assessed as some concerns or high risk of bias);
- Inconsistency (i.e. wide variance of point estimates across studies, minimal overlap of CIs, or a combination);
- Indirectness;
- Imprecision (i.e. small study, defined as fewer than 400 participants);
- Publication bias.

Based on the GRADE assessment, we determined the level of certainty (high, moderate, low, or very low) that the estimates of the effect were correct. Where synthesis was not possible due to a lack of studies, we applied GRADE to individual studies. We documented our decisions to downgrade the certainty of the evidence for the above criteria using footnotes.

#### *Methods for future updates*

##### Living systematic review considerations

Whilst we will continue to conduct monthly searches and immediately extract data, assess risk of bias and incorporate new evidence into the syntheses as soon as it becomes available, we will wait until the accumulating evidence changes one or more of the following components of the review before republishing the review:

- critical outcome findings;
- credibility (e.g. GRADE rating) of the critical outcomes;
- new settings, populations, interventions, or comparisons studied.

We will review the review scope and methods approximately yearly, or more frequently if appropriate, in light of potential changes in the topic area or the evidence being included in the review. We will consider the necessity for the review to be living each year against the required criteria for living systematic reviews, including whether the uncertainty in the evidence is ongoing; whether further relevant research is likely; and by assessing the ongoing relevance of the question to decision-makers.

### Consumer involvement

The review question and outcomes were developed in consultation with stakeholders, including consumers, health professionals (e.g. community health service managers), and policymakers (e.g. New South Wales Ministry of Health), to ensure that the findings of the review will be used to guide decision-making on the types of interventions that should be funded to address e-cigarette use in children and adolescents. The author team consulted with consumers during the development of the review protocol to ensure the scope, aims, and methods of the review met the needs of end-users in order to maximise impact.

## RESULTS

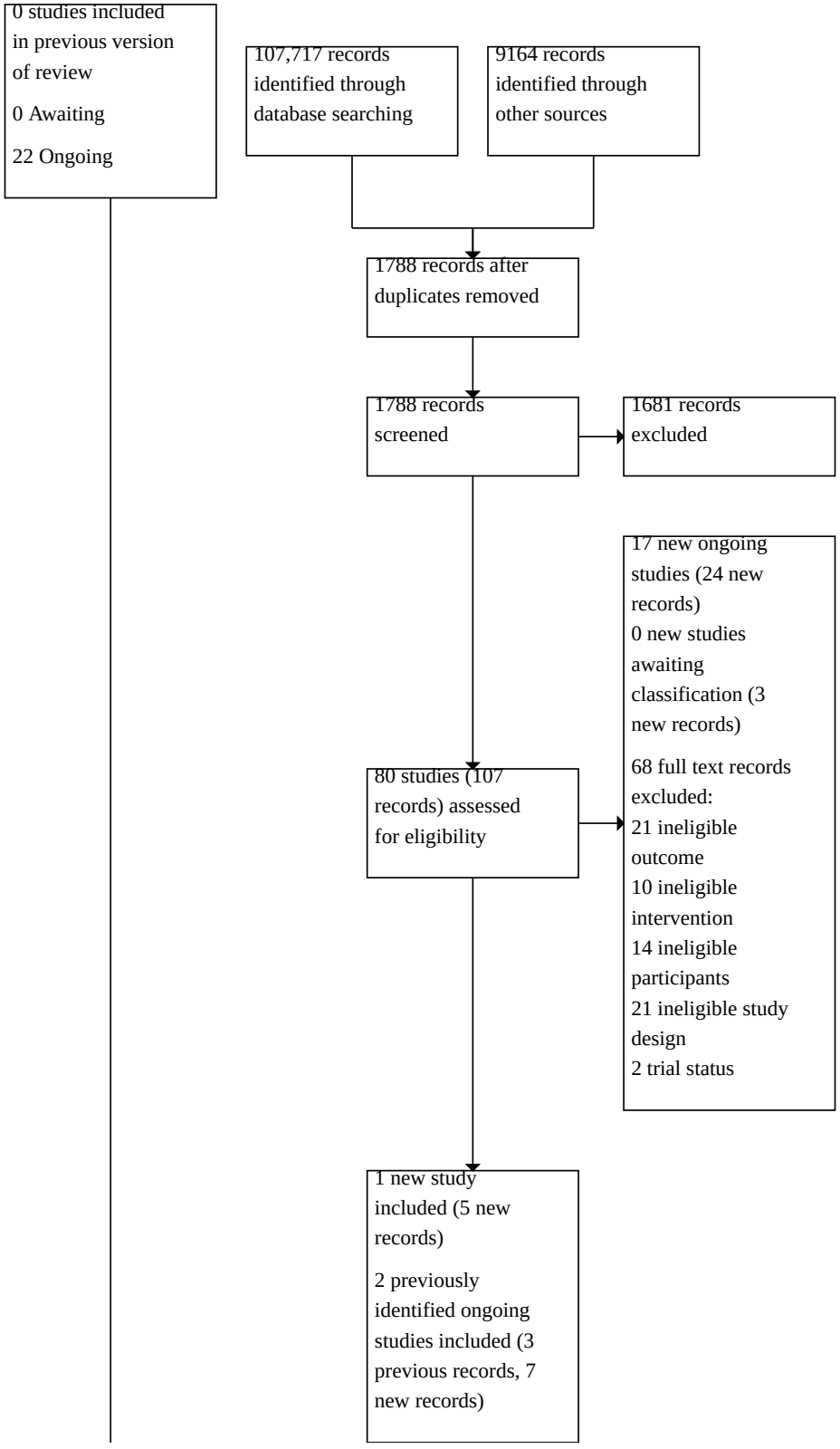
### Description of studies

#### Results of the search

For this update, the search of electronic databases identified 107,717 records (see Figure 1 for PRISMA flow diagram) and 9164 records were identified from other sources, resulting in 1788 records following de-duplication (4294 total, including 1251 from the previous review). No additional records were identified from contacting authors of ongoing studies, including those with published protocols, or from handsearching the reference lists of ongoing studies. Following screening of titles and abstracts, we obtained the full texts of 80 studies (107 records) for further review (190 studies total, including 110 from the previous review). Three studies (reported in 15 records) met the inclusion criteria for the review.


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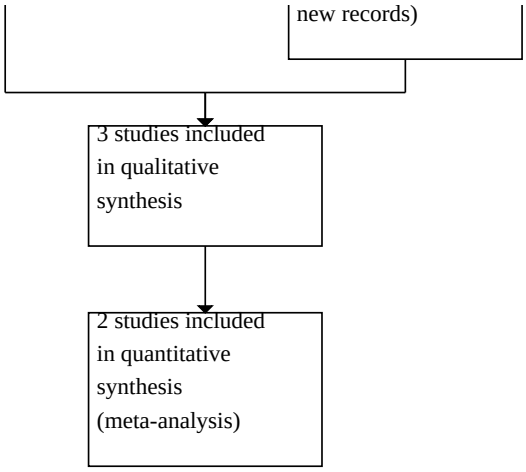
**Figure 1. Study flow diagram.**





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**Figure 1. (Continued)**




We identified 17 new studies (24 records) as ongoing (30 studies total, including 13 from the previous review) (see Supplementary material 5). We identified three new records that belong to studies previously identified as awaiting classification in the previous version of the review, but no new studies. There are five studies total (all 5 identified in the previous review) (see Supplementary material 4) that may be eligible for inclusion in a future review update.

### Included studies

The three included studies (Graham 2024 [77, 78, 79, 80]; Beeres 2022 [81, 82, 83, 84, 85]; Gardner 2025) have been summarised below and in the Characteristics of Included Studies tables (Supplementary material 2).

#### *Study design*

Of the three included studies, two studies (Beeres 2022; Gardner 2025) employed a cluster-RCT design with two study arms (intervention; and minimal intervention or usual practice control). The remaining study (Graham 2024) employed an RCT design with three study arms (intervention; assessment-only control; and waiting-list control) (Table 1).

#### *Participants and settings*

Two studies (Beeres 2022; Gardner 2025) were conducted in the school setting. One school-based study (Beeres 2022) was conducted in schools located in Sweden and included 1176 participants (aged approximately 13 years) from 43 clusters (i.e. schools), whilst the remaining school-based study (Gardner 2025) was conducted in Australia and included 7653 participants (aged approximately 12–14 years) from 49 randomised schools. The final study (Graham 2024) was community-based, conducted in the US, and included 1681 participants aged 13–17 years who were current users of e-cigarettes.

#### *Interventions*

Two prevention studies tested school-based interventions targeting the prevention of adolescent e-cigarette use (Beeres 2022; Gardner 2025). For one study (Beeres 2022), adolescents received the 3-year Tobacco-free Duo (T-Duo) intervention consisting of a signed agreement with a significant adult ('caregiver') pledging to remain tobacco-free throughout their compulsory schooling. Adolescents and their caregiver attended an information session in the first year of the intervention, and adolescents also received classroom education consisting of age-specific information and practical activities four times per year for the length of the intervention. For the remaining school-based study (Gardner 2025), adolescents received the OurFutures vaping prevention programme, consisting of four 40-minute lessons comprising information about e-cigarettes and tobacco and assertive refusal strategies, delivered approximately one week apart.

One cessation study (Graham 2024) tested a 14-week intervention targeting e-cigarette cessation amongst adolescents. Adolescents randomly allocated to receive the intervention received a tailored text-message-based programme for four weeks (those without a quit date) or 14 weeks (those with a quit date). The text messages contained content on a range of topics, including mental health support and the use of nicotine replacement therapy in consultation with a healthcare professional.

#### *Comparators*

For one school-based prevention study (Beeres 2022), adolescents allocated to the minimal intervention control group received the classroom education component of the T-Duo programme only, consisting of four classroom lessons per year for the length of the intervention. For the remaining school-based prevention study (Gardner 2025), adolescents allocated to the active control group received the usual health education curriculum. For the cessation study (Graham 2024), adolescents were randomly allocated to the assessment-only control, where they received monthly text-messages asking to report e-cigarette use up to the 6-month follow-up, or a waiting-list control. All control participants in this study were provided with instructions on how to enrol in the 'This is Quitting programme' after the final follow-up period.

#### *Outcomes*

In relation to the critical outcomes of our review, one prevention study (Beeres 2022) reported on never use of e-cigarettes at 18–


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21 months post-baseline. We inverted this outcome to provide data on ever use as per our critical review outcome. The remaining prevention study (Gardner 2025) reported on ever use (among never-users at baseline) and past 30-day use of e-cigarettes (i.e. current use) at 12-month follow-up. The cessation study (Graham 2024) reported 30-day point prevalence abstinence from e-cigarettes at seven months follow-up and repeat 30-day point prevalence abstinence from e-cigarettes at one and seven months follow-up. All included studies assessed the critical review outcomes using self-report measures (e.g. surveys).

For the important review outcomes, one prevention study (Beeres 2022) reported on never use of sinus (Swedish tobacco leaf product), cigarettes and waterpipes at 18–21 months post-baseline, which were all inverted to ever use. The cessation study (Graham 2024) reported on 30-day use of combustible tobacco products at seven months follow-up. Additionally, both prevention studies (Beeres 2022; Gardner 2025) intended to report undesirable side effects from the intervention. One study (Gardner 2025) reported no adverse events, whilst for the remaining study (Beeres 2022), no data have been reported at the time of publication of this review update. Self-reported measures were used by all included studies to assess the important outcomes.

#### *Funding*

All included studies reported funding sources. One study (Beeres 2022) was funded by the Public Health Agency of Sweden, one study (Gardner 2025) was funded by the Medical Research Future Fund and the Australian National Health and Medical Research Council, whilst the remaining study (Graham 2024) was funded by the Truth Initiative.

### Excluded studies

Of the 80 studies (107 records) evaluated at the full-text review stage (190 total, including 110 from the previous review), 62 were considered to be ineligible (150 total, including 88 from the previous review). The primary reasons for studies to be excluded from this review update included: ineligible outcome (19 studies: 7 studies did not include an eligible measure of e-cigarette use; and 12 studies had a follow-up of fewer than 3 months;); ineligible intervention (9 studies: all 9 studies did not include a specific intervention component targeting e-cigarettes); ineligible study design (21 studies: all 21 studies were not RCTs); ineligible participants (11 studies: all 11 studies included participants with a mean age of over 19 years at baseline) and trial status (e.g. trial abandoned due to lack of funding) (2 studies, 2 records). A subset of key excluded studies have been reported in the characteristics of excluded studies table (Supplementary material 3), with a full list of excluded studies available on request.

## Risk of bias in included studies

Risk of bias assessments for the critical review outcomes are reported in the risk of bias tables for interventions to prevent e-cigarette use (Analysis 1.1; Analysis 2.1) and interventions to cease e-cigarette use (Analysis 4.1). Risk of bias assessments for important review outcomes are reported in Supplementary material 6.

For the critical review outcome of e-cigarette ever use, two prevention studies (Beeres 2022; Gardner 2025) employed cluster-RCT designs to investigate an intervention to prevent e-cigarette use. Beeres 2022 was judged as having some concerns overall, with low risk for randomisation and timing of recruitment of participants (clustered trials) and missing outcome data, but having some concerns for deviations from intended interventions, outcome measurement and selective reporting. Gardner 2025 was judged as having high risk overall due to timing of the recruitment of participants (clustered trials), missing outcome data, and selective reporting.

For the critical review outcome of e-cigarette current use, one prevention study (Gardner 2025) was judged as having high risk overall due to timing of the recruitment of participants (clustered trials) and having some concerns for randomisation and outcome measurement. The cessation study (Graham 2024), which employed an RCT design to investigate an intervention to cease e-cigarette use, was judged to have a low risk of bias for all domains for the e-cigarette current use outcome.

## Synthesis of results

See Summary of findings 1; Summary of findings 2; and Supplementary material 7.

### Comparison 1: Interventions to prevent e-cigarette use versus control

#### *Critical outcomes*

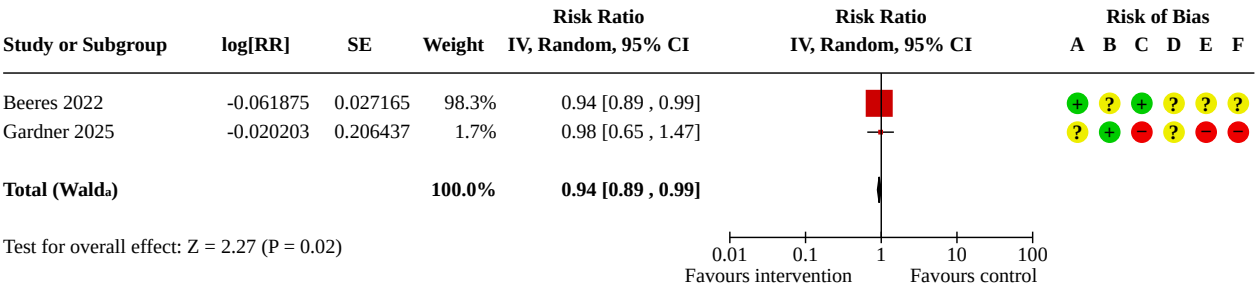
See Summary of findings 1

##### *Ever use of e-cigarettes*

Two studies (Beeres 2022; Gardner 2025) examined the effect of prevention-focused interventions on adolescent ever-use of e-cigarettes and were pooled within a meta-analysis. Findings of the meta-analysis indicated that these interventions may be effective in preventing adolescent e-cigarette ever-use relative to usual care, although this evidence is very uncertain (RR 0.94, 95% CI 0.89 to 0.99; 2 studies; 5306 participants; random-effects model; very low-certainty evidence; Analysis 1.1; Figure 2).


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**Figure 2.**




Test for overall effect: Z = 2.27 (P = 0.02)

Heterogeneity: Tau² (REML[b]) = 0.00; Chi² = 0.04, df = 1 (P = 0.84); I² = 0%

**Footnotes**

[a]CI calculated by Wald-type method.

[b]Tau² calculated by Restricted Maximum-Likelihood method.

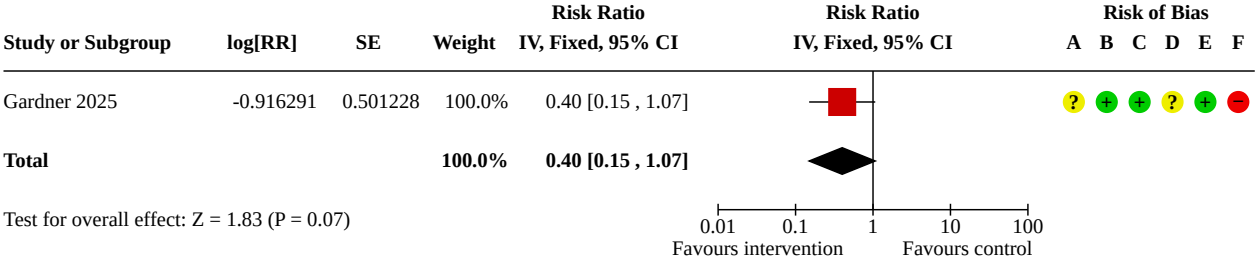
**Risk of bias legend**

(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

***Current use of e-cigarettes***

One study (Gardner 2025) examined the effect of a prevention-focused intervention which may have little to no difference on adolescent current use of e-cigarettes, although this evidence is very uncertain (RR 0.40, 95% CI 0.15 to 1.07; 1 study; 3949 participants; very low-certainty evidence; Analysis 2.1; Figure 3).

**Figure 3.**




Test for overall effect: Z = 1.83 (P = 0.07)

Heterogeneity: Not applicable

**Risk of bias legend**

(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

***Important outcomes***

***Ever use of tobacco***

One study (Beeres 2022) reported that a prevention-focused intervention may have little to no difference in preventing adolescent ever use of tobacco products, including: cigarettes (RR 0.97, 95% CI 0.93 to 1.02; 1 study; 1398 participants; Analysis 3.1); smoking a whole cigarette (RR 0.96, 95% CI 0.87 to 1.06; 1 study; 1438 participants; Analysis 3.2); snus (RR 0.96, 95% CI 0.86 to 1.07; 1 study; 1423 participants; Analysis 3.3); and waterpipes (RR 0.97, 95% CI 0.88 to 1.07; 1 study; 1394 participants; Analysis 3.4).

***Current use of tobacco***

No studies reported this outcome.


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#### *Unintended adverse consequences of the intervention*

Gardner 2025 reported no adverse effects.

### Comparison 2: Interventions to cease e-cigarette use versus control

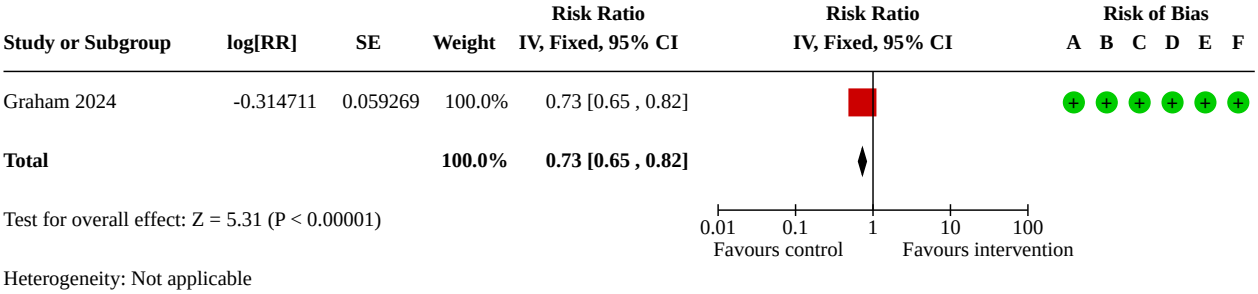
#### *Critical outcomes*

See Summary of findings 2.

#### *Current use of e-cigarettes*

One study (Graham 2024) found that a cessation-focused intervention is likely to reduce adolescent current use of e-cigarettes (RR 0.73, 95% CI 0.65 to 0.82; 1 study, 1064 participants; moderate-certainty evidence; Analysis 4.1; Figure 4).

**Figure 4.**




**Risk of bias legend**
(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

#### *Important outcomes*

##### *Ever use of tobacco*

No studies reported this outcome.

##### *Current use of tobacco*

One study (Graham 2024) found that a cessation-focused intervention may reduce adolescent current use of combustible tobacco products (CTP) amongst dual users at baseline (RR 0.73, 95% CI 0.57 to 0.93; 1 study; 418 participants; Analysis 5.1), but may have little to no difference amongst exclusive e-cigarette users (RR 0.76, 95% CI 0.50 to 1.14; 1 study; 598 participants; Analysis 5.2).

##### *Unintended adverse consequences of the intervention*

No studies reported this outcome.

### Equity assessment

All included studies (Beeres 2022; Graham 2024; Gardner 2025) reported data on a number of PROGRESS-Plus characteristics, including adolescent age, gender and sex, and education. The average age of participants ranged from 13.30 years for Gardner 2025, through to 16.4 years for Graham 2024. The average age of participants in Beeres 2022 was not reported, although all participants were in Year 7 when recruited so were likely to be aged 12–13 years. Gender was evenly balanced in Beeres 2022 (approximately 50% female, 48% male, and 2% other), and Gardner 2025 (approximately 46% female, 51% male, and 3% gender diverse, prefer not to say or missing). There was a lower proportion of male participants in Graham 2024 (approximately 50% female, 42% male, and 8% non-binary or other). All participants were in Year 7 (Beeres 2022) or Year 7 or 8 (Gardner 2025) when recruited, whereas most (approximately 75%) participants in Graham 2024 were in Year 11 or 12 when recruited. One study (Gardner 2025) reported on adolescent race and ethnicity and socio-educational advantage, whilst another study (Beeres 2022) reported caregiver race and ethnicity, occupation and socioeconomic status. No studies reported adolescent social capital, or religion, and no studies reported conducting subgroup analyses by any PROGRESS-Plus characteristics.

### Reporting biases

Not applicable.

## DISCUSSION

### Summary of main results

This review sought to assess the effectiveness of interventions to prevent or cease e-cigarette use in children and adolescents (aged 19 years and younger). Despite employing a comprehensive search strategy, only three studies were identified as eligible for inclusion in the review. Two studies (Beeres 2022; Gardner 2025) investigated the effectiveness of school-based interventions to prevent adolescent e-cigarette use, whilst the remaining study (Graham 2024) investigated the effectiveness of a community-based intervention to cease adolescent e-cigarette use. Findings from the included studies indicate that interventions to prevent adolescent electronic cigarette use (very low-certainty evidence), and those aimed at cessation of adolescent electronic cigarette use


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(moderate-certainty evidence), may be effective. As the findings are based on only a few studies, and the effects remain uncertain, they should be interpreted with caution. Given the number of ongoing studies, we anticipate the number of eligible studies included in the review will rapidly accumulate and conclusions regarding effectiveness may change.

For the critical review outcome of adolescent ever use of e-cigarettes, meta-analysis of the two prevention studies showed that school-based interventions may be effective in preventing uptake, although this evidence is very uncertain (Beeres 2022; Gardner 2025). Whilst one study (Gardner 2025) reported there may be little to no difference between adolescent ever or current use, findings from the remaining study (Beeres 2022) suggest that school-based programmes, consisting of intervention components targeting adolescents (i.e. curriculum activities) and parents (e.g. parent-adolescent education sessions; pledge to remain e-cigarette free), may be an effective approach for preventing adolescent ever use of e-cigarettes (Analysis 1.1; Analysis 2.1). Little to no difference from this study was observed in preventing adolescent ever use of other tobacco products (i.e. a review important outcome), including cigarettes, snus and waterpipes (Analysis 3.1; Analysis 3.2; Analysis 3.3; Analysis 3.4). The findings are in contrast with previous systematic reviews that have found that multi-component school-based interventions targeting the school environment, parents and students, are effective in improving other student health behaviours, including nutrition and tobacco uptake [86]. However, the effects of multi-component school-based interventions on student e-cigarette use were not evaluated in these reviews.

Regarding current adolescent use of e-cigarettes (i.e. a critical review outcome), our review finds that community-based interventions, such as tailored text-message programmes sent to adolescents, are likely to be an effective public health approach to reduce adolescent e-cigarette use amongst current users (Analysis 4.1; Analysis 4.2). For our secondary important outcome, positive effects were also observed for the reduction of current use of combustible tobacco products amongst adolescents who were dual users of e-cigarettes and combustible tobacco (Analysis 5.1), but not for those who were exclusive e-cigarette users at baseline (Analysis 5.2). This is also broadly consistent with previous systematic reviews reporting that text-message-based interventions can be effective in improving other adolescent health behaviours, including smoking cessation [87].

### Limitations of the evidence included in the review

As only three studies met the review eligibility criteria, the generalisability of the review findings are limited. Additionally, the certainty of the evidence for the critical review outcome of adolescent e-cigarette use was judged as very low-to-moderate, reducing confidence in the effect estimates. The limited number of included studies and heterogeneity in study type (i.e. two prevention-focused studies; one cessation-focused study) and outcome measures (i.e. ever use and current use) precluded pooling in additional meta-analyses as previously planned in the review protocol. The small number of studies also precluded the conduct of any subgroup or specificity analyses to determine if the overall effect remained robust under certain conditions.

Several factors may have contributed to the limited number of included studies in this review update. First, youth e-cigarette use is an emerging public health issue and, as such, research into this population is still in its infancy. This is evident in the results of our search strategy, with more than 75% of citations included in title and abstract screening published in the past five years. Second, the stringent review eligibility criteria, including the types of eligible study designs and outcomes measures, likely precluded the inclusion of numerous studies. For example, the potential impact of e-cigarette policy and regulatory changes (e.g. Tobacco 21 laws), including restrictions on e-cigarette sales and marketing, on adolescent e-cigarette use are often evaluated as part of natural experiments, not RCTs (i.e. the eligible study design) [88, 89, 90]. Further, numerous studies were excluded for solely examining the effect of interventions on child or adolescent susceptibility to e-cigarette use, not ever or current use (i.e. the review critical outcome) [91, 92, 93]. Other studies were excluded for using a follow-up period of less than three months. Such studies may reflect the challenges of undertaking trials of prevention interventions, which need to accommodate long follow-up periods given the latency of the targeted behaviour change.

### Limitations of the review processes

We undertook a comprehensive search for studies, including searching electronic databases in addition to trial registries, with no restrictions on publication date or language of publication. It is possible that the search strategy did not capture unregistered and/or unpublished trials in the grey literature. Additionally, the limited number of included studies (n = 3) led to changes in the review methods compared to what was previously published in the review protocol, particularly regarding the measurement of the treatment effects and synthesis approach. These changes have been documented in the 'Differences between protocol and review' section.

### Agreements and disagreements with other studies or reviews

The limited and uncertain evidence on the effect of interventions to prevent or cease adolescent e-cigarette use is consistent with the findings of recent reviews. For example, a systematic review of 39 studies conducted by Mylocopos and colleagues [94] concluded that some school-based programmes show promise for preventing e-cigarette initiation among children and young people. However, most of the school-based programmes were evaluated using non-randomised study designs, including cross-sectional and pre-post studies, and thus causation could not be established. Additionally, only a limited number of studies examined adolescent ever or current use of e-cigarettes, with the majority instead solely assessing intentions and attitudes towards e-cigarettes. This is similar to a previous overview of adolescent e-cigarette prevention programmes conducted by Liu and colleagues [48], who determined that such interventions have yielded encouraging results, but few had empirically evaluated the impact on adolescent e-cigarette use. Additionally, whilst Heshmati and colleagues [95] identified seven RCTs aimed at evaluating interventions to cease e-cigarette use, only one of the included interventions targeted adolescents.

Our findings are also broadly consistent with a recent systematic review of randomised and non-randomised studies that examined the efficacy of school-based e-cigarette preventive interventions [38]. Whilst the meta-analysis of five studies (including Beeres 2022 from our review) found no evidence to suggest school-based programmes were effective in preventing e-cigarette use at the


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longest follow-up, subgroup analyses indicated that results varied based on follow-up time point. For example, the overall effect became significant when studies with follow-up < 12 months were excluded [38]. Schools have long been recommended as a crucial setting for the implementation of health promotion programmes targeting a range of child and adolescent health behaviours, more recently including e-cigarette use. However, further rigorous evaluations of school-based e-cigarette prevention programmes developed to specifically address factors associated with adolescent e-cigarette use are required to strengthen the evidence-base [43].

Further, Graham 2024 was recently included in a meta-analysis that aimed to synthesise the results from RCTs of interventions to cease e-cigarette use [95]. Three of the six included studies (including Graham 2024) used digital interventions, such as smartphone applications and text-messages, to support cessation. Findings of the meta-analyses demonstrated an increase in the odds of achieving 7-day PPA and continuous abstinence following intervention [95]. However, given the mean age of participants ranged from 20.3 to 52.6 years of age, the applicability of these findings for the development of adolescent-specific e-cigarette cessation programmes may be limited. As such, there is a paucity of evidence to support the use of text-message programmes, or other types of programmes, to help support adolescents to cease e-cigarette use.

## AUTHORS' CONCLUSIONS

### Implications for practice

The review findings suggest that school-based health promotion programmes may prevent the uptake of e-cigarettes, whilst community-based programmes (e.g. text messages) show promise in improving adolescent quitting. However, the evidence base (n = 3), provides a limited basis to guide decisions of policymakers and practitioners on the types of health promotion programmes or other activities that should be implemented to support the prevention or cessation of adolescent e-cigarette use. In the absence of such evidence, policymakers and practitioners remain reliant on more formative evidence (e.g. non-randomised, pre-test/post-test studies) in addition to existing guidance frameworks and consensus statements, such as the WHO Framework Convention on Tobacco Control, to inform decision-making and practice [25]. The framework, which has recently been applied to e-cigarettes, provides guidance to assist with country-level implementation of interventions to reduce the demand for tobacco and e-cigarettes [26].

#### Equity-related implications for practice

Interventions to prevent or cease child and adolescent e-cigarette use may help address health disparities if they are designed and implemented to reach populations at greater risk, such as those from lower socioeconomic backgrounds, rural or remote areas, or marginalised communities. However, evidence within this review provides limited guidance for equity-related practice, as neither of the included studies specifically examined intervention effects by health-equity factors, such as the PROGRESS-Plus Characteristics. Current findings of the review regarding prevention are more applicable to younger adolescents, whilst cessation findings are more applicable to older adolescents. In the absence of more robust evidence, practitioners and policymakers must rely heavily on local knowledge and stakeholder engagement, particularly with adolescents and their communities, to help ensure interventions are contextually appropriate and feasible to implement.

### Implications for research

Given the limited evidence on the effectiveness of interventions to prevent or cease child and adolescent e-cigarette use, any research in this space that employs high-quality study designs, would be of considerable value. This includes additional school- and community-based interventions to strengthen the evidence reported in this review update. Reporting of adverse effects was also commonly missing. Increased reporting of this will aid in the reduction of this evidence gap. Promisingly, the number of ongoing studies suggest that such evidence will soon rapidly emerge. Ongoing surveillance of this evidence has been recommended [96]. Specifically, living systematic reviews are recommended in these instances where there is a high level of uncertainty in the existing evidence that will impact on the conclusions of the review [49]. The review will continue to be undertaken as a living systematic review given its strong alignment to required criteria for living systematic reviews.

#### Equity-related implications for research

Future research should consider how health-related equity factors, such as the PROGRESS-Plus characteristics (e.g. socioeconomic status, geographic location, ethnicity and gender) influence the effectiveness and reach of interventions to prevent and/or cease adolescent e-cigarette use. Studies should aim to include diverse populations and report outcomes by relevant subgroups to identify differential effects and ensure that interventions do not unintentionally widen existing health disparities.

## SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: 10.1002/14651858.CD015511.pub3.

**Supplementary material 1** Search strategies

**Supplementary material 2** Characteristics of included studies

**Supplementary material 3** Characteristics of excluded studies

**Supplementary material 4** Characteristics of studies awaiting classification

**Supplementary material 5** Characteristics of ongoing studies

**Supplementary material 6** Risk of bias

**Supplementary material 7** Analyses

**Supplementary material 8** Data package

## ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions:

Cochrane People, Health Systems and Public Health Thematic Group and Cochrane Public Health supported the authors in the development of this review.

The following people conducted the editorial process for this article:

• Sign-off Editor (final editorial decision): Toby Lasserson, Deputy Editor-in-Chief;

• Managing Editor (provided editorial guidance to authors, edited the article, and collated peer-reviewer comments ): Marwah Anas El-Wegoud, Cochrane Central Editorial Service;

• Editorial Assistant (conducted editorial policy checks, selected peer reviewers, and supported the editorial team): Lisa Wydrzynski, Cochrane Central Editorial Service;

• Copy Editor (copy editing and production): Anne Lethaby, Cochrane Central Production Service;

• Peer-reviewers (provided comments and recommended an editorial decision): Richard van Zyl-Smit, University of Cape Town, South Africa ( (clinical/content review); Adele Susan Feeney, DNP, FNP-C Associate Professor and Associate Dean of Advanced Practice Programs Tan Chingfen Graduate School of Nursing at UMass Chan Medical School (clinical/content review); Jennifer Pearson, School of Public Health, University of Nevada, Reno, USA (clinical/content review); Maciej Sumara (patient and public review); Clare Miles, Evidence Production and Methods Directorate (methods review); Steve McDonald, Cochrane Australia (search review).

## Contributions of authors

CBa, LW, and SLY conceived the idea for this review. CBa, LW, SLY, RH, AH, and ES contributed to the methods for the protocol. CBa, CBi and HT developed the search strategy for the review. CBa, RH, HT, SMc, CBi, SLY, DDA and SMa screened titles and abstracts and determined study eligibility. SMc and SLY conducted risk of bias assessment. ES, RH and CBa conducted GRADE assessment. HT and CBi entered data into RevMan. CBa, HT, and LW led the drafting of the manuscript. All authors provided support for manuscript drafting and provided critical comments and final approval for the manuscript.

## Declarations of interest

If a review author was also a contributor to a study that may be of interest to the review, that review author was not involved in the determination of eligibility, data extraction, risk of bias assessment, and GRADE assessment. In such instances, these roles were carried out by review authors who were not directly involved in these studies, in line with the Cochrane Conflict of Interests policy.

CBa: reports that she is an affiliated member of the Hunter Medical Research Institute.

HT: no interest declared.

SMc: SMc is an Executive Member of Cochrane Public Health and the Nutrition and Public Health Thematic Group and the People, Health Systems, and Public Health Thematic Group, but has not been involved in the editorial process or assessment of this review. She has no interest to declare.

RH: reports that they are employed as a Service Manager at NSW Health responsible for the implementation of health promotion programmes to children aged five to 18 years; they are an affiliated member of the Hunter Medical Research Institute.

SLY: reports she has published opinions in medical journals relevant to the interventions in the work. No other interest declared.

ES: reports that she has previously been contracted (paid to her institution) by the Australian Department of Health to provide a report on e-cigarette use and trends in Australia to advise government policy, as well as contracted (paid to her institution) by the New South Wales Department of Health to assist in developing educational resources for young people regarding the risks of e-cigarette use. ES has also published opinions in medical journals relevant to the interventions in the work. ES is a co-author of an included study in the review (Gardner 2025) and a partially credited inventor of the programme detailed in the study.

AH: reports that she is an Editor for the Australian New Zealand Journal of Public Health for which she receives an honorarium. AH is an affiliated member of the Hunter Medical Research Institute and a member of the Public Health Association of Australia.

CBi: no interests declared.

SMa: no interests declared.

DDA: no interest declared.

LW: reports that he is the Co-ordinating Editor of Cochrane Public Health and is an Executive Member of the Nutrition and Public Health Thematic Group and the People, Health Systems, and Public Health Thematic Group, but has not been involved in the editorial process or assessment of this review. LW is an affiliated member of the Hunter Medical Research Institute.

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## What's new

| Date | Event | Description |
|---|---|---|
| 30 July 2026 | Amended | The plain language summary has been updated to correct a typo in the main results section |

## History

Protocol first published: Issue 11, 2022
Review first published: Issue 11, 2023

| Date | Event | Description |
|---|---|---|
| 3 July 2026 | New search has been performed | Search updated |
| 3 July 2026 | New citation required and conclusions have changed | Three studies were included in this update (two prevention studies, one cessation study). A meta-analysis was conducted for the ever-use outcome in prevention studies; findings from the cessation study were summarised narratively. No changes were made to the review question. Minor methodological and wording updates were made to improve clarity and align with current *Cochrane Handbook* guidance. |
| 1 January 2024 | Amended | Plain language summary title added |

## Registration and protocol

This review is registered in the Cochrane Database of Systematic Reviews. The protocol (https://doi.org/10.1002/14651858.CD015511) and previous version of the review (https://doi.org/10.1002/14651858.CD015511.pub2) are published in the Cochrane Library.

## Data, code and other materials

All data extracted from included studies, along with risk of bias assessments, are available within RevMan Web (Supplementary material 8). Search strategies are provided in Supplementary material 1. No additional analytic code was used beyond the built-in functions of RevMan Web. Data extraction forms and other supporting materials are available from the corresponding author upon reasonable request.


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## ADDITIONAL TABLES

**Table 1. Overview of Synthesis and Included Studies table**

| Study ID | Country of conduct | Age of participants | Study design | Study arms | Number randomised (intervention/control) | Outcomes reported | Time point measured |
|---|---|---|---|---|---|---|---|
| *Intervention Type: Prevention* | | | | | | | |
| Beeres 2022 | Sweden | Grade 7 (~13 years old) | Cluster-RCT | 1. Intervention: T-Duo programme and caregiver intervention<br>2. Control: Only T-Duo programme | *Schools:* 43 (intervention: 21, control: 22)<br>*Students:* 1176 (intervention: 840, control: 936) | **Critical outcome relating to never-users (e-cigarettes):** Never use of e-cigarettes (student self-report)<br>**Important outcome relating to child and adolescent use of tobacco:** Never use of tobacco (%) (student self-report), never smoked a whole cigarette (%) (student self-report), never use of snus (%) (student self-report), never use of waterpipes (%) (student self-report)<br>**Important outcome relating to reported adverse events:** Undesirable side effects of the full-component intervention | 9 months (T1), 18-21 months (T2) |
| Gardner 2025 | Australia | Grade 7-8 (12-14 years) | Cluster-RCT | 1. Intervention: OurFutures Vaping programme<br>2. Control: Usual health education lessons | *Schools:* 49 (intervention: 26, control: 23)<br>*Students:* 7653 (intervention: 3836, control: 3817) | **Critical outcome relating to never-users (e-cigarettes):** Ever use of e-cigarettes among never-users at baseline (%) (student self-report)<br>**Critical outcome relating to current-users (e-cigarettes):** Past 30-day e-cigarette use (%) (student self-report) | 5 weeks, 6 months, 12 months |
| *Intervention Type: Cessation* | | | | | | | |
| Graham 2024 | USA | 13–17-year-olds | RCT | 1. Intervention: Text messages with tailored support<br>2. Assessment only control: Vaping assess- | 1681 (intervention: 759, assessment only control: 744, waiting-list control: 178) | **Critical outcome relating to current-users (e-cigarettes):**<br>30-day point prevalence of abstinence from e-cigarettes (%) (student self-report) and repeated point prevalence of abstinence from e-cigarettes (student self-report)<br>**Important outcome relating to child and adolescent use of tobacco:** 30-day use of | 1 month, 7 months |


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**Table 1. Overview of Synthesis and Included Studies table** *(Continued)*

| | | | | |
|---|---|---|---|---|
| | | | ment text messages<br>3. Text messages following final follow-up | combustible tobacco products (of exclusive e-cig users at baseline) (%) (student self-report) and 30-day use of combustible tobacco products (of dual users of combustible tobacco products and e-cigarettes at baseline) (%) (student self-report) |


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